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DUAL FUNCTION EDGE DEVICE AND METHOD FOR ACCELERATING UE-SPECIFIC BEAMFORMING

Abstract

An edge device includes an antenna array and control circuitry. The control circuitry tracks a position of user equipment (UE) in motion in terms of a current distance and a current angle from the edge device and executes beamforming to communicate a beam of radio frequency (RF) signal to the UE. The beam of RF signal is communicated in a specific radiation pattern based on the track of the position of the UE. Also, the beam of RF signal has a signal strength greater than a threshold. The control circuitry further calibrates the specific radiation pattern of the antenna array to further communicate the beam of RF signal in a specific beam shape that is suitable for a defined period.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS/INCORPORATION BY REFERENCE
[0001] This patent application makes reference to, claims priority to, claims the benefit of, and is a Continuation application of U.S. patent application Ser. No. 18/397,427, filed on Dec. 27, 2023, which is a Continuation application of U.S. Pat. No. 11,899,123, issued on Feb. 13, 2024, which is a Continuation application of U.S. Pat. No. 11,874,389, issued on Jan. 16, 2024, which is a Continuation application of U.S. Pat. No. 11,550,019, issued on Jan. 10, 2023, which is a Continuation application of U.S. Pat. No. 11,366,195, issued on Jun. 21, 2022, which is a Continuation application of U.S. Pat. No. 11,275,147 issued on Mar. 15, 2022. [0002] This application further makes reference to U.S. application Ser. No. 17/341,978, filed on Jun. 8, 2021. [0003] Each of the above referenced applications are hereby incorporated herein by reference in their entirety.

FIELD OF TECHNOLOGY

[0004] Certain embodiments of the disclosure relate to wireless communication. More specifically, certain embodiments of the disclosure relate to a dual function edge device and a method for accelerating user equipment (UE) specific beamforming.

BACKGROUND

[0005] Wireless telecommunication in modern times has witnessed the advent of various signal transmission techniques and methods, such as beamforming and beam steering techniques, for enhancing the capacity of radio channels. Latency and the high volume of data processing are considered prominent issues with next-generation networks, such as 5G. Currently, the use of edge computing in the next generation networks, such as 5G and upcoming 6G, is an active area of research, and many benefits have been proposed, for example, faster communication between vehicles, pedestrians, and infrastructure and other communication devices. For example, it is proposed that proximity of conventional edge devices to user equipment (UEs) may likely reduce the response delay usually suffered by UEs while accessing the traditional cloud. However, there are many open technical challenges for successful and practical use of edge computing in modern networks, especially in 5G or the upcoming 6G environment.

[0006] In a first example, one major technical challenge of the mmWave beamforming is signal attenuation, which adversely impacts low latency and high data rate requirements. For example, generally, mmWave signals may be easily blocked by rain or absorbed by oxygen, which is one reason why it only works at short ranges. Unlike traditional antennas that broadcast in every direction, so other communication devices can wirelessly connect with them, 5G-enabled antennas do not broadcast but points a beam at one object and may make an individual connection to one or more objects. This increases the complexity of antennas in user equipment (UE), base stations, and

other network nodes (e.g., repeater devices, small cell, etc.) as antennas are required to be designed to handle the complexity of aiming a beam at a target object in a crowded cellular environment with plenty of obstructions. Current positioning methods used to determine a geographical location of a target device, such as a UE, are coarse (having more than 3 to 10 meters error) and add to the ever-increasing signaling load among various network nodes to estimate position. For example, in 3GPP release 16, it is planned to achieve less than 3 meters positioning accuracy for some use cases. In certain scenarios, the complexity increases manifold when the target object is in motion and its location changes rapidly. Thus, in such scenarios, faster decisions to alter the beam become necessary to ensure the best performance. Moreover, the performance of UEs varies with the location of the UEs and their proximity to a relay or service side of a conventional repeater. This is because the usual method of wide beam access, although it works in proximity to the conventional repeater device but suffers as a given UE, moves at greater distances from the conventional repeater device, especially for mmWave communication due to signal attenuation.

[0007] In a second example, Quality of Experience (QoE) is another open issue, which is a measure of a user's holistic satisfaction level with a service provider (e.g., Internet access, phone call, or other carrier network-enabled services). The challenge is how to ensure seamless connectivity as well as QoE without significantly increasing infrastructure cost, which may be commercially unsustainable with present solutions.

[0008] Further limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art through comparison of such systems with some aspects of the present disclosure as set forth in the remainder of the present application with reference to the drawings.

BRIEF SUMMARY OF THE DISCLOSURE

[0009] A dual function edge device and a method for accelerating user equipment (UE) specific beamforming for high performance and reliable communication, substantially as shown in and/or described in connection with at least one of the figures, as set forth more completely in the claims.

[0010] These and other advantages, aspects and novel features of the present disclosure, as well as details of an illustrated embodiment thereof, will be more fully understood from the following description and drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is a network environment diagram illustrating various components of an exemplary communication system with a central cloud server and a plurality of edge devices for accelerating user equipment (UE) specific beamforming, in accordance with an exemplary embodiment of the disclosure.

[0012] FIG. 2A is a block diagram illustrating different components of an exemplary edge device for accelerating user equipment (UE) specific beamforming, in accordance with an embodiment of the disclosure.

[0013] FIG. 2B is an illustration of an antenna array of a dual-function edge device, in accordance with an embodiment of the disclosure.

[0014] FIG. 2C is an illustration of an antenna array of a dual-function edge device, in accordance with another embodiment of the disclosure.

[0015] FIG. 3 is a first exemplary scenario for implementation of the edge device and method for accelerating user equipment (UE) specific beamforming for high performance and reliable communication, in accordance with an embodiment of the disclosure.

[0016] FIGS. 4A, 4B, and 4C, collectively, is a flowchart that illustrates an exemplary method for accelerating user equipment (UE) specific beamforming for high performance and reliable

communication, in accordance with an embodiment of the disclosure.

[0017] FIG. 5 is a flowchart that illustrates an exemplary method for accelerating user equipment (UE) specific beamforming for high performance and reliable communication, in accordance with another embodiment of the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0018] Certain embodiments of the disclosure may be found in a dual function edge device and a method for accelerating user equipment (UE) specific beamforming for high performance and reliable communication. Beneficially, the edge device has a dual function capability of sensing and beamforming that work in synchronization such that a beam of radio frequency (RF) signal having a signal strength greater than a threshold can be concentrated on at least the UE even if the UE is in motion and changes its location rapidly. The disclosed dual function edge device executes sensing function and beamforming function cooperatively in real-time or near real-time that enables making faster and accurate decisions to alter the beams as per need without any increase in signaling load on a cellular network and further ensures the best performance consistently in terms of high throughput data rate as well as ultra-reliable communication as compared to existing systems. Further, the edge device and the method of the present disclosure ensure seamless connectivity as well as QoE while reducing the infrastructure cost due to effective management and concentration of radiation pattern of the beams of RF signals having higher signal strength to higher distances due to precise sensing and beamforming functions that work in cooperation. For instance, existing road-side units (RSU) or repeater devices need to be deployed near each other (say X meters apart) to provide coverage in an area due to the issues of signal attenuation in mmWave communication. This is because the usual method of wide beam access, although it works in proximity to the conventional repeater device but suffers as a given UE, moves at greater distances from the conventional repeater device, especially for mmWave communication due to signal attenuation. On the contrary, the use of the disclosed dual-function edge device reduces the number of such devices that need to be deployed to provide coverage in the same area by more than 30-50 percent. This is because of the cooperative operation of the sensing function and the beamforming function that seamless connectivity with higher signal strength is provided to the moving UE by dynamic and accurate directing of the beam of RF signal in a specific radiation pattern to the UE. In the following description, reference is made to the accompanying drawings, which form a part hereof, and in which is shown, by way of illustration, various embodiments of the present disclosure.

[0019] FIG. 1 is a network environment diagram illustrating various components of an exemplary communication system with a central cloud server and a plurality of edge devices, in accordance with an exemplary embodiment of the disclosure. With reference to FIG. 1, there is shown a block diagram **100** of a network environment that includes a central cloud server **102**, a plurality of edge devices **104**, one or more user equipment (UEs) **106**, and a plurality of base stations **108**. There is further shown a plurality of different WCNs **110**, such as a first WCN **110A** of a first service provider and a second WCN **110B** of a second service provider.

[0020] The central cloud server **102** includes suitable logic, circuitry, and interfaces that may be configured to communicate with the plurality of edge devices **104**, the one or more UEs **106**, and the plurality of base stations **108**. In an example, the central cloud server **102** may be a remote management server that is managed by a third party different from the service providers associated with the plurality of different WCNs **110**. In another example, the central cloud server **102** may be a remote management server or a data center that is managed by a third party, or jointly managed, or managed in coordination and association with one or more of the plurality of different WCNs **110**. In an implementation, the central cloud server **102** may be a master cloud server or a master machine that is a part of a data center that controls an array of other cloud servers communicatively coupled to it for load balancing, running customized applications, and efficient data management. The connectivity enhanced database **102A** may be a low-latency database, for example,

“DynamoDB,” “Scylla,” or other proven and known low-latency databases that can handle one or more million transactions per second on a single cloud server. The connectivity enhanced database **102A** may be an intelligent and learned database, which specifies wireless connectivity enhanced information for a surrounding area of each of the plurality of edge devices **104** independent of a plurality of different WCN **110** of different service providers. An exemplary training phase to generate the connectivity enhanced database **102A** has been provided and explained in detail in the U.S. application Ser. No. 17/341,978, filed on Jun. 8, 2021.

[0021] Each edge device of the plurality of edge devices **104** includes suitable logic, circuitry, and interfaces that may be configured to communicate with the central cloud server **102**. Each edge device of the plurality of edge devices **104** such as the edge device **104A** may be one of an XG-enabled edge communication device, an XG-enabled edge repeater device, an XG-enabled relay device, an XG-enabled customer premise equipment (CPE), an XG-enabled small cell, or an XG-enabled road-side unit (RSU), where the term “XG” refers to 5G or 6G communication. In an example, the edge device **104A** may be controlled out-of-band, for example, in a management plane, by the central cloud server **102**. In an implementation, some of the edge devices of the plurality of edge devices **104** may be deployed at a fixed location, while some may be portable. For example, an edge device may be a fixed wireless access (FWA) device, a repeater device, a small cell, or a CPE or a portable repeater device.

[0022] Each of one or more UEs **106**, such as a first UE **106A** and a second UE **106B**, may correspond to telecommunication hardware used by an end-user to facilitate communications. Alternatively stated, each of the one or more UEs **106** may refer to a combination of a mobile equipment and subscriber identity module (SIM). Each of the one or more UEs **106** may be a subscriber of at least one of the plurality of different WCNs **110**. Examples of the one or more UEs **106** may include, but are not limited to, a smartphone, a vehicle, a virtual reality headset, an augmented reality device, an in-vehicle device, a wireless modem, or any other customized hardware for telecommunication.

[0023] Each of the plurality of base stations **108** may be a fixed point of communication that may communicate information, in the form of a plurality of beams of RF signals, to and from communication devices, such as the one or more UEs **106** and the plurality of edge devices **104**. Multiple base stations corresponding to one service provider may be geographically positioned to cover specific geographical areas. In an implementation, each of the plurality of base stations **108** may be a gNB. In another implementation, the plurality of base stations **108** may include eNBs, Master eNBs (MeNBs) (for non-standalone mode), and gNBs.

[0024] Each of the plurality of different WCNs **110** is owned, managed, or associated with a mobile network operator (MNO), also referred to as a mobile carrier, a cellular company, or a wireless service provider that provides services, such as voice, SMS, MMS, Web access, data services, and the like, to its subscribers, over a licensed radio spectrum. Each of the plurality of different WCNs **110** may own or control elements of network infrastructure to provide services to its subscribers over the licensed spectrum, for example, 4G LTE or 5G spectrum (FR1 or FR2). For example, the first base station **108A** or the first UE **106A** may be controlled, managed, or associated with the first WCN **110A**, and the second base station **108B** or the second UE **106B** may be controlled, managed, or associated with the second WCN **110B** different from the first WCN **110A**. The plurality of different WCNs **110** may also include mobile virtual network operators (MVNO).

[0025] Beneficially, the plurality of edge devices **104** provides full coverage in 5G NR cellular network deployments without any adverse user experience to meet QoS for 5G NR communication. Moreover, the central cloud server **102** and the plurality of edge devices **104** exhibit a decentralized model that not only brings cloud computing capabilities closer to UEs to reduce latency but also manifests several known benefits for various service providers associated with the plurality of different WCNs **110**. For example, backhaul traffic is reduced by provisioning content at the edge, computational resources are distributed geographically in different locations (e.g., on-premises

mini cloud, central offices, customer premises, etc.) depending on the use case requirements and reliability of a network is improved by distributing content between edge devices and the centralized cloud server **102**. Apart from these and other known benefits (or inherent properties) of edge computing, the central cloud server **102** and the plurality of edge devices **104** improve and solve many open issues related to the convergence of edge computing and modern wireless networks, such as 5G or upcoming 6G. The central cloud server **102** significantly improves the beam management mechanism of 5G new radio (NR), true 5G, and creates a platform for upcoming 6G communications, to achieve low latency and high data rate requirements.

[0026] The plurality of edge devices **104**, such as the edge device **104A**, significantly reduces the issue of signal attenuation, which adversely impacts low latency and high data rate requirements. The edge device **104A** is configured to perform continuous tracking of a position of a UE in motion, such as the first UE **106A**, accurately with a centimeter-level accuracy (e.g., less than 5-10 cm error magnitude) instead of a coarse level accuracy in existing systems. Accordingly, a radiation pattern of an antenna array of the edge device **104A** may be calibrated to communicate a beam of RF signal in a specific radiation pattern that is most suited for a defined period. In an example, if the first UE **106A** is moving away from the edge device **106A**, a pencil beam may be employed for communication having signal strength greater than the threshold so that both uplink and downlink communication may be maintained (or increase) at the multigigabit data rate. The radiation pattern may be dynamically updated based on the ongoing changes in the position of the first UE **106A** as it moves. Such concentration of the beam of RF signal having increased signal strength may be executed for the one or more UEs **106** that are recognized by the edge device **104A** as valid devices to receive such services from the edge device **104A** for consistent high-performance and ultra-reliable communication. As the beam of RF signal is in a concentrated or targeted form and directed precisely on the ongoing tracked position, mmWave communication occurs at extended ranges than usual short ranges. Furthermore, the antenna array (e.g., a first antenna array **210A** of FIG. 2A) of the edge device **104A** is designed and configured to reduce the complexity of aiming a beam at a target object in a crowded cellular environment with plenty of obstructions, without increasing any signaling load among various network nodes to estimate the position of the one or more UEs **106** even when in motion. The edge device **104A** may be a dual-function edge device that ensures faster decisions due to integrated sensing and calibration functions to alter the beam direction, angle, and shape dynamically and rapidly to ensure the best performance and QoE by providing seamless connectivity without increasing infrastructure cost. Furthermore, in some implementations, the edge device **104A** uses the same antenna array for sensing as well as calibration of the radiation pattern, which not only ensures a compact design of antenna and the edge device **104A** but also reduces (i.e., optimizes) the power usage of the edge device **104A**.

[0027] FIG. 2A is a block diagram illustrating different components of an exemplary edge device, in accordance with an embodiment of the disclosure. FIG. 2A is explained in conjunction with elements from FIG. 1. With reference to FIG. 2A, there is shown a block diagram **200** of the edge device **104A** with various components. The edge device **104A** is one of the plurality of edge devices **104** (FIG. 1). The edge device **104A** has a donor side **202A** facing towards the plurality of base stations **108**, such as the first base station **108A** and the second base station **108B** (of FIG. 1). The edge device **104A** also has a service side **202B** facing towards the one or more UEs **106**, such as the first UE **106A**. In an implementation, the edge device **104A** may include a control section **204** and a front-end radio frequency (RF) section, which may include one or more donor antenna arrays **206** and an uplink chain **208** at the donor side **202A**, and further one or more service antenna arrays **210** and a downlink chain **212** at the service side **202B**. The control section **204** may be communicatively coupled to the front-end RF section, such as the one or more donor antenna arrays **206**, the uplink chain **208**, the one or more service antenna arrays **210**, and the downlink chain **212**. The front-end RF section supports millimeter-wave (mmWave) communication as well communication at a sub 6 gigahertz (GHz) frequency. The control section **204** may further include

control circuitry **214** and a memory **216**. There is further shown a sensing function (SF) **218** and a beamforming function (BF) **220**. In some implementation, the edge device **104A** may further include a sensing radar **222** and a radar data memory **224** that are communicatively coupled to the control circuitry via a Serial Peripheral Interface (SPI) **226**. In some implementations, a separate sensing radar, such as the sensing radar **222** and the radar data memory **224**, may not be provided, and a portion of a service antenna array, such as the first antenna array **210A**, of the one or more service antenna arrays **210** may be configured for sensing a surrounding area of the edge device **104A**.

[0028] The edge device **104A** includes suitable logic, circuitry, and interfaces that may be configured to communicate with one or more base stations of the plurality of base stations **108**, one or more UEs **106**, and the central cloud server **102**. The edge device **104A** may be further configured to communicate with the one or more UEs **106** and other edge devices of the plurality of edge devices **104**. In accordance with an embodiment, the edge device **104A** may support multiple and a wide range of frequency spectrum, for example, 3G, 4G, 5G, and 6G (including out-of-band frequencies). The edge device **104A** may be at least one of an XG-enabled repeater device, an XG-enabled small cell, an XG-enabled road-side unit (RSU), an XG-enabled relay device, an XG-enabled vehicle-mounted edge device, where the term “XG” refers to 5G or 6G radio communication. Other examples of the edge device **104A** may include, but is not limited to, a 5G wireless access point, an evolved-universal terrestrial radio access-new radio (NR) dual connectivity (EN-DC) device, a Multiple-input and multiple-output (MIMO)-capable repeater device, or a combination thereof deployed at a fixed location.

[0029] The one or more donor antenna arrays **206** may be provided at the donor side **202A** of the edge device **104A** and may be communicatively coupled to an uplink chain **208**. The one or more service antenna arrays **210** may be provided at the service side **202B** and may be communicatively coupled to the downlink chain **212**. Each of the uplink chain **208** and the downlink chain **212** may include a transceiver chain, for example, a cascading receiver chain and a cascading transmitter chain, each of which comprises various components for baseband signal processing or digital signal processing. For example, the cascading receiver chain various components, such as a set of low noise amplifiers (LNA), a set of receiver front end phase shifters, and a set of power combiners, for the signal reception (not shown here for brevity). Similarly, the cascading transmitter chain comprises various components for baseband signal processing or digital signal processing, such as a set of power dividers, a set of phase shifters, a set of power amplifiers (PA).

[0030] In an implementation, the one or more service antenna arrays **210** at the service side **202B** may be configured to execute mmWave communication with the one or more UEs **106** (including vehicles) within its communication range. In an implementation, the one or more service antenna arrays **210** also supports multiple-input multiple-output (MIMO) operations and may be configured to execute MIMO communication with the one or more UEs **106** within its communication range. The MIMO communication may be executed at a sub 6 gigahertz (GHz) frequency or at mmWave frequency for 5G NR communication. Each of the one or more donor antenna arrays **206** and the one or more service antenna arrays **210** may be one of an XG phased-array antenna panel, an XG-enabled antenna chipset, an XG-enabled patch antenna array, or an XG-enabled servo-driven antenna array, where the “XG” refers to 5G or 6G. Examples of implementations of the XG phased-array antenna panel include, but are not limited to, a linear phased array antenna, a planar phased array antenna, a frequency scanning phased array antenna, a dynamic phased array antenna, and a passive phased array antenna.

[0031] The control circuitry **214** may be communicatively coupled to the memory **216** and the front-end RF section. The control circuitry **214** may be configured to execute various operations of the edge device **104A**. The control circuitry **214** may be configured to control various components of the front-end RF section, such as the one or more donor antenna arrays **206** and the uplink chain **208** at the donor side **202A**; and the one or more service antenna arrays **210** and the downlink chain

212 at the service side **202B**. The edge device **104A** may be a programmable device, where the control circuitry **214** may execute instructions stored in the memory **216**. Examples of the implementation of the control circuitry **214** may include but are not limited to an embedded processor, a baseband processor, a Field Programmable Gate Array (FPGA), a microcontroller, a specialized digital signal processor (DSP), a control chip, a Reduced Instruction Set Computing (RISC) processor, an Application-Specific Integrated Circuit (ASIC) processor, a Complex Instruction Set Computing (CISC) processor, and/or other processors, or state machines.

[0032] The memory **216** may be configured to store values calculated by the control circuitry **214**. Examples of the implementation of the memory **216** may include, but are not limited to, a random access memory (RAM), a dynamic random access memory (DRAM), a static random access memory (SRAM), a processor cache, a thyristor random access memory (T-RAM), a zero-capacitor random access memory (Z-RAM), a read-only memory (ROM), a hard disk drive (HDD), a secure digital (SD) card, a flash drive, cache memory, and/or other non-volatile memory. It is to be understood by a person having ordinary skill in the art that the control section **204** may further include one or more other components, such as an analog to digital converter (ADC), a digital to analog (DAC) converter, a cellular modem, and the like, known in the art, which are omitted for brevity.

[0033] In an implementation, the sensing function **218** may be a function, such as a feature, which, when activated, the sensing of objects in the surrounding area of the edge device **104A** is initiated. In an example, the sensing function **218** may be implemented as a sensing circuit that may be activated and deactivated. For example, when deactivated, the edge device **104A** may be in a power-saving mode.

[0034] In an implementation, the beamforming function **220** may be a function, such as a feature, used to execute beamforming to direct one or more beams of RF signals to one or more target objects in the surrounding area in one or more specific radiation patterns. In an example, the beamforming function **220** may be activated and deactivated. For example, when deactivated, the edge device **104A** may be in a power-saving mode.

[0035] In operation, the control circuitry **214** may be configured to activate a sensing function **218** to sense a surrounding area of the edge device **104A**. The sensing may be executed by communicating a radio frequency (RF) wave that hits the surrounding objects of the edge device **104A** and detects the energy that is reflected from such objects. The RF wave may be in mmWave signal frequency or an out-of-band frequency that is different from the frequency used to communicate data streams in the cellular network of the plurality of different WCNs **110**.

[0036] In an implementation, the RF wave may be a chirp signal with a starting frequency in the mmWave frequency range. The chirp signal may have a plurality of chirp parameters. In an example, the plurality of chirp parameters may include a start frequency of about 60 Gigahertz (GHz), a slope of about 70 Megahertz per unit sample (MHz/us), samples per chirp of about 256 samples, chirps per frame of about 32, a sampling rate of about 5.2 mega samples per second (Msps), sweep bandwidth of about 3.44 GHz, and frame periodicity of about 250 milliseconds (msec).

[0037] The control circuitry **214** may be further configured to generate a three-dimensional (3D) environment representation of the surrounding area of the edge device **104A**. In an example, the 3D environmental representation may be generated in the form of a point cloud. The 3D environment representation may be generated based on the sensing of the surrounding area of the edge device **104A**. The 3D environmental representation may be a representation of a surrounding environment of the edge device **104A** that indicates a plurality of mobile and stationary objects surrounding the edge device **104A**. For example, the 3D environmental representation may indicate any possibility of signal blockages or fading, road condition, traffic information, and current weather condition.

[0038] The control circuitry **214** may be further configured to recognize a first UE **106A** in motion to be a valid device to receive one or more services from the edge device **104A**. The edge device

104A may be in an offline mode or a connected mode. The connected mode refers to a setting in which the edge device **104A** may be communicatively coupled to the central cloud server **102**. The offline mode refers to a setting or mode where the edge device **104A** may not be communicatively coupled to the central cloud server **102**. In both the offline mode and the connected mode, the control circuitry **214** can recognize the first UE **106A** in motion to be the valid device.

[0039] In an implementation, when the first UE **106A** is in the communication range of the edge device **104A**, an authentication key may be automatically communicated by the first UE **106A** to the edge device **104A**. The authentication key may be validated locally by the edge device **104A** in the offline mode or via the central cloud server **102** in the connected mode, and if the validation is successful, the first UE **106A** may be recognized as the valid device. In another implementation, the first UE **106A** may have an application that controls the validation between the first UE **106A** and the edge device **104A**, for example, based on a registered gesture or defined validation data, such as a hash value. In an example, the first UE **106A** may be a smartphone that may have the application installed in it, where the application may be communicatively coupled to the central cloud server **102**. In another example, the first UE **106A** may be a vehicle, in which the application may be installed in a smartphone connected to an in-vehicle infotainment system of the vehicle, or the application may be preinstalled in the vehicle (e.g., in the in-vehicle infotainment system). A unique identity, for example, in the form of the authentication key, or the registered gesture, or other identifying means may be used to identify users associated with the first UE **106A** as a valid user to receive services of the central cloud server **102** and the edge device **104A**. In yet another implementation, the control circuitry **214** may be configured to receive a connection request from the first UE **106A** associated with the first WCN **110A** of a first service provider. The connection request may be received via an out-of-band communication, such as Wi-Fi, Bluetooth, Li-Fi, a sidelink request (e.g., LTE sidelink, 5G New Radio (NR) sidelink, NR C-V2X sidelink), a vehicle-to-infrastructure (V2I) request, a personal area network (PAN) connection, or other out-of-band connection requests. The control circuitry **214** may be further configured to validate the first UE **106A** based on the connection request.

[0040] In some implementations, in the connected mode, the validation of the first UE **106A** as the valid device by the edge device **104A** may be controlled by the central cloud server **102** based on communication of sensing information, for example, a current position, a moving direction, a time-of-day, a device ID associated with the first UE **106A**. The central cloud server **102** may identify that the device ID is known and communicate such information to the edge device **104A** in advance before such first UE **106A** reaches the communication range of the edge device **104A** as the central cloud server **102** has information of the travel path of the first UE **106A** based on the sensing information received from the first UE **106A**. Thus, the control circuitry **214** can recognize the first UE **106A** in motion to be the valid device directly or indirectly with the assistance of the central cloud server **102**.

[0041] The control circuitry **214** may be further configured to track a position of the first UE **106A** in motion from the edge device **104A**. Based on the continuous sensing, the control circuitry **214** may track the position of the first UE **106A** in motion from the edge device **104A**. Such tracking may be a local tracking using the communicated RF wave emitted by the edge device **104A** that detects the first UE **106A** in terms of a current distance and a current angle from the edge device **104A**. As the location coordinates of the edge device **104A** are known, the location coordinates of the first UE **106A** may be derived without increasing any signaling load on the network. In a case where the communicated RF wave is emitted in a mmWave frequency, the accuracy of tracking is increased due to the high sensitivity of the mmWave frequency to motion. Moreover, the tracking is insensitive to low light or weather and manifest a centimeter-level accuracy in the tracking of the position of the first UE **106A**. In some implementations, the tracking of the position of the first UE **106A** in motion may be further enhanced by correlating the tracking information with the generated 3D environment representation of the surrounding area of the edge device **104A**.

[0042] The control circuitry **214** may be further configured to execute beamforming to direct a first beam of RF signal having a signal strength greater than a threshold to the first UE **106A** in motion that is recognized as the valid device, based on the track of the position of the first UE **106A** and the generated 3D environment representation. In an implementation, the beamforming function **220** may be a function, such as a feature, which when activated, the first antenna array **210A** directs one or more beams of RF signals to one or more target objects in the surrounding area in one or more specific radiation patterns. Based on the continuous track of the position of the first UE **106A**, its position is accurately determined with centimeter-level accuracy (e.g., less than 5-10 cm level accuracy) instead of a coarse level accuracy in existing systems. Further, the generated 3D environment representation provides a reference to correlate the tracked position of the moving object, i.e., the first UE **106A**, with a set of the point cloud that belongs to the same moving object, i.e., the first UE **106A**. This correlation further increases the tracking efficiency where the rapidly changing position of the first UE **106A** can be tracked with improved accuracy. Accordingly, the radiation pattern of the first antenna array **210A** may be calibrated to communicate the first beam of RF signal in a specific beam shape that is most suited for a defined period, for example, for next “X” seconds, where “X” is a positive integer, such as 5 sec, 10 sec, 15 sec, etc. In an example, if the first UE **106A** is moving away from the edge device **104A**, a pencil beam or a narrow beam with a beam angle greater than the pencil beam may be employed for communication of the first beam of RF signal with the signal strength greater than the threshold so that both uplink and downlink communication may be maintained (or increase) at the multigigabit data rate. The radiation pattern may be dynamically updated based on the changes in the position of the first UE **106A**. Such concentration of the beam of RF signal having increased signal strength may be executed for devices that are recognized by the edge device **104A** as a valid device to receive such service of consistent high-performance and ultra-reliable communication.

[0043] In one implementation, the edge device **104A** may be a dual-function edge device, where a first portion of the first antenna array **210A** is used for the sensing and one or more second portions of the first antenna array are used for the beamforming to direct the first beam of RF signal in a defined radiation pattern. Advantageously, in such an implementation, a portion of the same phase array antenna, such as the first antenna array **210A**, may be used for the sensing, such as tracking the position of the first UE **106A**, whereas other portions of the same phase array antenna may be used for communication of the first beam of RF signal having the signal strength greater than the threshold to the first UE **106A** in motion. The use of a portion of the first antenna array **210A** for sensing not only ensures a compact design of antenna and the edge device **104A** but also reduces (i.e., optimizes) the power usage of the edge device **104A**. As compared to the use of a separate sensing radar, such as the sensing radar **222**, which uses additional power to operate, the use of the same antenna for the dual function of the sensing and beamforming significantly improves (i.e., optimizes) the power usage of the edge device **104A** to perform both the sensing and beamforming. Moreover, since the first antenna array **210A** act as a common point of reference to determine both the position of the first UE **106A** as well as to communicate the first beam of RF signal to the first UE **106A**, the chances of error are reduced, and the sensing and beamforming may be cooperatively executed in real-time or near real-time for faster and accurate decisions to alter the beams of RF signals as per need without any increase in signaling load on a cellular network, such as the first WCN **110A** or the second WCN **110B** as compared to existing systems.

[0044] Alternatively, in another implementation, the edge device **104A** may be a dual-function edge device that comprises the sensing radar **222** communicatively coupled to the control circuitry **214**, and where the sensing function **218** may be activated in the sensing radar **222** to sense the surrounding area of the edge device **104A**. Examples of the sensing radar **222** may include, but may not be limited to, Frequency-Modulated Continuous Wave (FMCW) radar, a Lidar, or other mmWave sensors. In this implementation, a separate sensing radar may be provided and integrated with the edge device **104A**. In such a case, the sensed data of the surrounding area of the edge

device **104A** may be stored at the radar data memory **224**. The sensing radar **222** may be communicatively coupled to the control circuitry **214** via the SPI **226**. The SPI **226** may be a full-duplex bus interface used to send data between the control section **204** (e.g., a microcontroller or DSP, such as the control circuitry **214**) and other peripheral components, such as the sensing radar **222** and a modem, for example, a 5G modem. The SPI **226** supports very high speeds and throughput and is suitable for handling a huge amount of data. In an example, the sensing radar **222** may be a mmWave sensor that manifests high sensitivity to motion, insensitivity to weather and low light, ability to operate at high velocity, and high-range resolution. For example, scene parameters of the sensing radar **222** may include a range resolution of about 4.4 centimeters (i.e., provides centimeter-level accuracy), maximum unambiguous range of about 9 meters, a maximum radial velocity of about 1 meter/sec, a radial velocity resolution of 0.12 meters/sec, and an azimuth resolution of about 14.5 degrees. It is to be understood that such scene parameters are for an exemplary implementation and may vary depending on other implementations; for example, the maximum unambiguous range may be more than 9 meters, such as 9-30 meters, and the like, without limiting the scope of the disclosure.

[0045] In accordance with an embodiment, the control circuitry **214** may be further configured to execute the sensing in a first frequency that is different from a second frequency used to communicate the first beam of RF signal in a defined radiation pattern. In other words, the communication of the RF wave for sensing purposes may be done in the first frequency, for example, 60 GHz or other frequency, which is different from the frequency of the first beam of RF signal that carries data stream to the first UE **106A**. The use of mutually isolated frequencies is employed to avoid any unwanted signal interference and signal attenuation. In one implementation the sensing may be processed by the same RF and base-band processor (such as the control circuitry **214**) which executes the beamforming. In another implementation the sensing and the beamforming operations only share the baseband processor or only RF circuits. In yet another implementation, each of the sensing function **218** (i.e., sensing of the surrounding area) and the beamforming function **220** (to execute beamforming) has its own radio and own baseband processor (i.e., a separate baseband processor or separate control circuitry).

[0046] In accordance with an embodiment, the control circuitry **214** may be further configured to determine location coordinates of a plurality of reflective objects in the surrounding area of the edge device **104A** and utilize the determined location coordinates of the plurality of reflective objects to correlate the radiation pattern of the first antenna array **210A** to the plurality of reflective objects for the calibration. Based on the sensing of the surrounding area of the edge device **104A**, the control circuitry **214** determines the location coordinates of the plurality of reflective objects that may cause RF signals from the edge device **104A** to reflect towards the edge device **104A**. Thus, if the location coordinates are known, the radiation pattern of the first beam of RF signal may be further configured to minimize any unwanted signal reflections from the plurality of reflective objects reducing signal noise.

[0047] In accordance with an embodiment, the control circuitry **214** may be further configured to determine a distance and an angle of the edge device **104A** from each of a plurality of mobile and stationary objects surrounding the edge device **104A**. Based on the sensing, once the distance and the angle of the edge device **104A** from each of the plurality of mobile objects, including the first UE **106A** and stationary objects surrounding the edge device **104A**, are determined, the radiation pattern of the first beam of RF signal may be further improved in terms of directivity and precision targeting of the highly concentrated first beam of RF signal with increased signal strength.

[0048] In accordance with an embodiment, the control circuitry **214** may be further configured to determine local traffic information in real-time or near real-time based on the sensed surrounding area of the edge device **104A** and the generated 3D environment representation of the surrounding area of the edge device **104A**. Many a time, traffic information from satellites (or maps) is not accurate as such traffic information depends on the number of users using such software

applications running map services. Based on the sensing, a number of objects, such as vehicles, moving in the surrounding area of the edge device **104A** may be counted, which in turn indicates traffic information accurately locally around the edge device **104A** independent of the use of the Internet-based map services. Furthermore, the correlation of such local traffic information with the generated 3D environment representation further improves the accuracy of the determination of the local traffic information in real-time or near real-time. Such local traffic information determined in real-time or near real-time may be then used to further improve radiation patterns to communicate one or multiple beams of RF signals concurrently for different UEs, such as the first UE **106A** and the second UE **106B**.

[0049] In accordance with an embodiment, the control circuitry **214** may be further configured to predict a start time and an end time of a signal blockage for the first UE **106A** in motion being serviced by the edge device **104A** from a second moving object based on a track of the second moving object in the surrounding area of the edge device **104A**. Dynamic nature of surroundings of the edge device **104A**, such as any change in surroundings, have the potential to adversely impact signal propagation, cause signal loss, poor reach, or signal blockage by an object, such as a moving object or a stationary object, in the surroundings. For example, in certain scenarios, there may be temporary signal blockage due to the second moving object that may block signals between the edge device **104A** and a base station, such as the first base station **108A** or there may be a blockage due to a non-line-of-sight (NLOS) between the first base station **108A** and the edge device **104A**. In some other scenarios, there may be signal blockage due to the second moving object that may block signals between the edge device **104A** and the first UE **106A**. Based on the sensing, for example, based on a determination of distance, angle, velocity, and moving direction of the first UE **106A** and the second moving object, with respect to the edge device **104A**, the control circuitry **214** may predict the start time and the end time of the signal blockage for the first UE **106A** in motion much before the signal blockage occurs. Thereafter, the control circuitry **214** may be further configured to communicate an alert of the predicted signal blockage for the first UE **106A** to the central cloud server **102** along with the predicted start time and the end time of the signal blockage to cause the central cloud server **102** to instruct another edge device of the plurality of edge devices **104** with specific initial access information to continue servicing the first UE **106A**. Alternatively, instead of the central cloud server **102**, the control circuitry **214** may be further configured to communicate an alert of the predicted signal blockage for the first UE **106A** to another edge device of the plurality of edge devices **104** along with at least the predicted start time of the signal blockage and a specific initial access information to cause the other edge device to continue servicing the first UE **106A**. Accordingly, the central cloud server **102** may be configured to select an appropriate edge device, such as the edge device **104B**, to communicate wireless connectivity enhanced information, including specific initial access information to the other edge device to bypass the initial access-search on the other edge device. Based on the specific initial access information, the other edge device may quickly switch over to the appropriate base station (e.g., using PCID of the base station and ARFCN) received from the central cloud server **102** to continue servicing the first UE **106A**. The specific initial access information may further indicate to select a particular service side beam index, e.g., a beam index #17 out of 0-63 and a particular radiation pattern to service the first UE **106A** bypassing the initial access search at the other edge device, such as the edge device **104B**, where the handover time is much lesser than the standard average mm-wave gNB handover time under same scenarios, such as same cell radius and traveling speed of the first UE **106A**.

[0050] In accordance with an embodiment, the control circuitry **214** may be further configured to recognize another UE, such as the second UE **106B**, in motion to be a valid device to receive one or more services from the edge device **104A** concomitantly with the first UE **106A**. The control circuitry **214** may be further configured to track a position of the second UE **106B** in motion concurrently with the first UE **106A** from the edge device **104A**. The control circuitry **214** may be

further configured to update the radiation pattern of the first antenna array **210A** such that a second beam of RF signal having the signal strength greater than the threshold is concentrated on the second UE **106B** in motion concomitant to the first beam of RF signal that is concentrated on the first UE **106A** in motion.

[0051] In accordance with an embodiment, the control circuitry **214** may be further configured to obtain a corresponding activation signal from each of the first UE **106A** and the second UE **106B**. The corresponding activation signal may be generated at the first UE **106A** and the second UE **106B** using at least one of an application installed in the first UE **106A** and the second UE **106B**, an authentication key, or a registered gesture. The control circuitry **214** may be further configured to activate the edge device **104A** to service the first UE **106A** and the second UE **106B** based on an authentication of the corresponding activation signal. In an example, the first UE **106A** or the second UE **106B** may be a smartphone or other UE that may have the application installed in it, where the application may be communicatively coupled to the central cloud server **102**. In another example, the first UE **106A** or the second UE **106B** may be a vehicle, in which the application may be installed in a smartphone connected to an in-vehicle infotainment system of the vehicle or the application may be preinstalled in the vehicle (e.g., in the in-vehicle infotainment system). A unique identity, for example, in the form of the authentication key, or the registered gesture, or other identifying means may be used to identify users associated with the first UE **106A** or **106B** as a valid user to receive services of the central cloud server **102** and the edge device **104A**. The control circuitry **214** may be further configured to activate the edge device **104A** to service the first UE **106A** and the second UE **106B** based on an authentication of the corresponding activation signal.

[0052] In accordance with an embodiment, the control circuitry **214** may be further configured to set an offline mode or a connected mode at the edge device **104A**. In the connected mode, the control circuitry **214** may be further configured to periodically communicate sensing information to the central cloud server **102** based on the sensed surrounding area of the edge device **104A**. The sensing information may comprise two or more of: a distance and an angle of the edge device **104A** from each of a plurality of objects surrounding the edge device **104A**, a position of the edge device **104A**, a location and a moving direction of a plurality of UEs including the first UE **106A**, a time-of-day, local traffic information, local road information, local construction information, a local traffic light information, and local weather information.

[0053] In accordance with an embodiment, the control circuitry **214** may be further configured to obtain wireless connectivity enhanced information from the central cloud server **102** based on a position of the edge device **104A** in the connected mode. The wireless connectivity enhanced information may include specific initial access information to bypass an initial access-search on the edge device **104A**. The wireless connectivity enhanced information provided by the central cloud server **102** for the first UE **106A** and the second UE **106B** may include selected initial access information to accelerate UE-specific beamforming for each of the first UE **106A** and the second UE **106B** and connectivity to one or more base stations, such as the first base station **108A** and the second base station **108B**, from the edge device **104A** bypassing an initial access-search on the edge device **104A**. The wireless connectivity enhanced information, including the specific initial access information for the edge device **104A**, is extracted from the connectivity enhanced database **102A**. The selected initial access information for the first UE **106A** and the second UE **106B** may indicate the best and/or optimal transmit (Tx) and receive (Rx) beam for the donor side **202A** and the best Tx-Rx beam for the service side **202B**, best Physical Cell Identities (PCIDs) associated with different service providers, best and/or optimal absolute radio-frequency channel number (ARFCNs), and a signal strength information associated with each of Tx beam and the Rx beam of the edge device **104A**. The best and/or optimal refers to wireless connectivity using multiple beams of RF signals at both the donor side **202A** and service side **202B** that has the highest signal strength (or among the top three available signal strengths), for example, for 5G NR signals and data

throughput rate higher than one or more specified threshold values, while executing uplink and downlink communication using one or more of the plurality of different WCNs **110**. The specific initial access information may indicate which beam index to set at an edge device, such as the edge device **104A**, for the uplink communication, a specific Physical Cell Identity (PCID) which indicates which gNB to connect to, or which WCN of the plurality of different WCNs **110** to select, which is the best absolute radio-frequency channel number (ARFCN), which specific beam configuration to set, or whether a connection to the base station is to be established directly or indirectly in an NLOS path using another edge device, such as the edge device **104B**, in a network of edge devices depending on the current location of the edge device **104A**. The specific initial access information may further indicate which beam index to set at an edge device, such as the edge device **104A** for the downlink communication, which WCN to select, which specific beam configuration to set, what power level of the RF signal may be sufficient, or an expected time period to service one or more UEs, such as the first UE **106A**, depending on the current location of the edge device **104A**.

[0054] The control circuitry **214** may be further configured to receive a response from the central cloud server **102** that no handover is required for the edge device **104A** for one or more of the plurality of UEs based on the communicated sensing information. In a case where a wireless connection (e.g., a cellular connectivity) of a UE that is in motion, such as the first UE **106A**, is about to become less than a threshold performance value, such performance drop may be predicted by the central cloud server **102** based on new sensing information received from one or more edge devices in the vicinity of the UE or from the first UE **106A** itself. For example, the first UE **106A** may be attached to the first base station **108A**, and as the first UE **106A** moves, the distance from the first base station **108A** may increase, and the signal strength may gradually decrease. Thus, based on the new sensing information, such as a moving direction of the first UE **106A**, a position of the first UE **106A**, distance from one or more edge devices in the vicinity of the first UE **106A**, a current weather condition, the location of the reflective objects around the first UE **106A**, and an overall 3D environment representation around the first UE **106A**, the central cloud server **102** determines that a handover is required to maintain QoE, and accordingly selects a suitable edge device (e.g., the edge device **104A** or the edge device **104B**) among the plurality of edge devices **104** and communicates wireless connectivity enhanced information to such selected edge device so that there is no need to perform beam sweeping operation or standard initial access search on such edge device. Thus, the first UE **106A** may readily connect to the edge device **104A** or another edge device, such as the edge device **104B**, and continue to perform uplink and downlink communication with high throughput without any interruptions. Similarly, in accordance with an embodiment, the central cloud server **102** may be further configured to determine that no handover is required for the edge device **104A** or the other edge device, such as the edge device **104B**, when a performance state of a wireless connection of the UE, such as the first UE **106A**, is greater than a threshold performance value.

[0055] Beneficially, multiple UEs, such as the first UE **106A** and the second UE **106B**, associated with different service providers may be serviced by a single edge device, such as the edge device **104A** (e.g., a repeater device or an RSU). Alternatively stated, a single edge device can service multiple UEs associated with different service providers, thereby reducing (i.e., optimizing) the total number of network nodes for each service provider required to be deployed or used to service the same number of UEs in one or more geographical areas. Furthermore, the edge device **104A** ensures seamless connectivity as well as QoE while reducing the infrastructure cost by greater than 50 percent considering just two different service providers, such as the first WCN **110A** and the second WCN **110B**. The edge device **104A** seamlessly handles heterogeneity and supports the plurality of different WCNs **110** for high performance and reliable communication. Furthermore, a consumer, such as the first UE **106A**, is provided to choose which WCN (i.e., which service provider) they like to connect to, and this is enabled from the cloud, such as the central cloud server

102. The central cloud server **102** transmits specific initial access information (optimal initial access information) associated with one or more WCNs, such as the first WCN **110A** and the second WCN **110B**, to the edge device **104A**, where such specific initial access information is used by the edge device **104A** to establish wireless connectivity bypassing conventional initial-access search. Hence, beneficially, a consumer of a UE, such as the first UE **106A**, subscribed to the first WCN **110A** can request the edge device **104A** in the connection request to relay an RF signal of the first WCN **110A**, and if the consumer of the first UE **106A** is subscribed to the second WCN **110B**, then the first UE **106A** can request the edge device **104A**, to relay an RF signal of the second WCN **110B**. Additionally, and advantageously, as the obtained connectivity enhanced information for the first UE **106A** and the second UE **106B** is independent of the plurality of different WCNs **110**, the complexity and the initial access latency is significantly reduced as the standard beam sweeping operation in the initial access phase is bypassed and is not required to be performed at the first UE **106A** and the second UE **106B** and the edge device **104A**, which in turn improves network performance and reduces additional signaling load (due to standard initial-access search) on associated WCNs of the plurality of different WCNs **110**.

[0056] In accordance with an embodiment, the control circuitry **214** is further configured to utilize the obtained connectivity enhanced information to, for example: reduce time to align to a timing offset of a beam reception at the edge device **104A** to a frame structure of a 5G NR radio frame, and allow uplink and downlink to use complete 5G NR frequency spectrum for different service providers; set beam index or set phase values of the received RF signals to design beams to service the first UE **106A** and the second UE **106B** in real time or near real time; set parameters, e.g., amplifier gains, and phase responses associated with the one or more donor antenna arrays **206** or the one or more service antenna arrays **210**; form specific beam patterns from the edge device **104A** specific to the first UE **106A** and the second UE **106B**; execute dynamic partitioning of a plurality of antenna elements of the first antenna array **210A** of the one or more service antenna arrays **210** at the service side **202B** into a plurality of spatially separated antenna sub-arrays to generate multiple beams in different directions at the same time or in a different time slot, where one of the partitions may be used for the sensing; set a suitable adjustment of a power back-off to minimize (i.e., substantially reduce) the impact of interference (echo or noise signals) and hence only use as much power as needed to achieve low error communication with the first base station **108A** or the second base station **108B** in the uplink or the first UE **106A** and the second UE **106B** in the downlink communication; and optimize blocks of radio and perform Radio access network optimization to improve coverage, capacity and service quality of a geographical area surrounding the edge device **104A**.

[0057] In another aspect, the edge device **104A** may comprise the first antenna array **210A** and the control circuitry **214**, where the control circuitry **214** may be further configured to activate a sensing function to sense a surrounding area of the edge device **104A**. The control circuitry **214** may be further configured to recognize the first UE **106A** in motion to be a valid device to receive one or more services from the edge device **104A**. The control circuitry **214** may be further configured to track a position of the first UE **106A** in motion from the edge device **104A**. The control circuitry **214** may be further configured to execute beamforming to direct a first beam of RF signal having a signal strength greater than a threshold to be concentrated on the first UE **106A** in motion that is recognized as the valid device, based on the track of the position of the first UE **106A**. The edge device **104A** may be a dual-function edge device in which a first portion of the first antenna array **210A** is used for the sensing and one or more second portions of the first antenna array **210A** is used for the beamforming to direct the first beam of RF signal in a specified radiation pattern.

[0058] FIG. 2B is an illustration of an antenna array of a dual-function edge device, in accordance with an embodiment of the disclosure. FIG. 2B is explained in conjunction with elements from FIGS. 1 and 2A. With reference to FIG. 2B, there is shown the first antenna array **210A** of the edge

device **104A** that may be a dual-function edge device, where a first portion **228** of the first antenna array **210A** is used for the sensing to sense a surrounding area of the edge device **104A** and a second portion **230** of the first antenna array **210A** used for the calibration for forming the first beam of RF signal in the calibrated radiation pattern. The first portion **228** may include a first set of antenna elements that radiate an RF wave in a first frequency (e.g., an out-of-band mmWave frequency) for the sensing of the surrounding area, such as to track the one or more UEs **106** in a first communication range. The second portion **230** may include a second set of antenna elements that radiate beamformed RF wave, i.e., the beam of RF signals in a second frequency (e.g., an in-band mmWave frequency that is owned or operated by one of the plurality of different WCNs **110**) for an uplink and a downlink cellular communication with the tracked one or more UEs **106** in a second communication range. In this implementation, the count of antenna elements in the second set of antenna elements may be greater than the first set of antenna elements. Furthermore, the first communication range may be greater than the second communication range so that sensing, tracking of the position of the one or more UEs **106**, such as the first UE **106A**, and the recognition of the one or more UEs **106**, such as the first UE **106A**, as a valid device may be successfully executed much ahead of time before the one or more UEs **106**, such as the first UE **106A**, enters the second communication range of the edge device **104A**. For example, the control circuitry **214** by use of the first portion **228** is configured to determine a velocity, a moving direction, a distance, an angle, a local travel path (e.g., a trajectory of motion) of the first UE **106A** based on the track of the first UE **106A** before the first UE **106A** enters the second communication range of the edge device **104A**. This provides the control circuitry **214** some leverage to perform the activation of the beamforming function **220** and perform preprocessing for the beamforming of the best suitable radiation pattern based on the tracking in advance so that as soon as the one or more UEs **106**, such as the first UE **106A**, enters the second communication range, the first beam of RF signal having a signal strength greater than a threshold can be concentrated on the first UE **106A** in motion that is already being tracked precisely and recognized as the valid device. The first beam of RF signal may relay a data stream to/from a source, such as the first base station **108A**, to/from the first UE **106A**.

[0059] In some implementations, the first portion **228** may be a predefined portion (in the first antenna array **210A**) that is configured for sensing. Similarly, the second portion **230** of the first antenna array **210A** may be another predefined portion (in the first antenna array **210A**) that is configured for the beamforming.

[0060] In some implementations, the control circuitry **214** may be configured to execute dynamic partitioning of a plurality of antenna elements of the first antenna array **210A** into a plurality of spatially separated antenna sub-arrays, where one spatially separated antenna sub-array may be used as a sensing radar for the sensing purpose whereas other spatially separated antenna sub-arrays may be used for the beamforming to either form one radiation pattern to communicate the first beam of RF signal calibration of multiple radiation patterns to generate multiple beams in different directions at the same time or in a different time slot. An example of dynamic partitioning of the plurality of antenna elements of the first antenna array **210A** is described in FIG. 2C.

[0061] FIG. 2C is an illustration of an antenna array of a dual-function edge device, in accordance with another embodiment of the disclosure. FIG. 2C is explained in conjunction with elements from FIGS. 1 and 2A. With reference to FIG. 2B, there is shown the first antenna array **210A** of the edge device **104A** that may be a dual-function edge device. In this implementation, the control circuitry **214** may be configured to execute dynamic partitioning of a plurality of antenna elements of the first antenna array **210A** into a plurality of spatially separated antenna sub-arrays. In this case, two portions of the first antenna array **210A**, such as portions **232A** and **232B**, may be configured to communicate two different beams of RF signals for two UEs, such as the first UE **106A** and the second UE **106B** concomitantly. The two different beams of RF signals may be communicated in different directions and angles and may have different radiation patterns, such as

one may be a narrow beam while the other may be a comparatively wider beam. Similarly, two portions of the first antenna array **210A**, such as portions **234A** and **234B**, may be configured to communicate two RF waves for sensing purposes to sense different areas (e.g., UEs coming towards edge device **104A** may be considered as one area, whereas UEs moving away from the edge device **104A** may be considered as another area) of the surrounding area of the edge device **104A** and track multiple UEs in motion at the same time.

[0062] FIG. **3** is an exemplary scenario for implementation of the edge device and the method for accelerating user equipment (UE)-specific beamforming for high-performance communication, in accordance with an embodiment of the disclosure. FIG. **3** is explained in conjunction with elements from FIGS. **1** and **2**. With reference to FIG. **3**, there is shown an exemplary scenario **300**. The exemplary scenario **300** includes a 5G-enabled repeater device, hereinafter simply referred to as a repeater device **302**, and user equipment (UE) **304**, which may be a vehicle in the exemplary scenario **300**. The repeater device **302** comprises the control circuitry **214**, a first antenna array **306** at a service side **302A** of the repeater device **302**, and a second antenna array **308** at a donor side **302B** of the repeater device **302**.

[0063] In accordance with the exemplary scenario **300**, the repeater device **302** corresponds to the repeater device **302**, and the UE **304** corresponds to the first UE **106A** (FIGS. **1** and **2**). The UE **304** may be a vehicle moving away from the repeater device **302** that may be deployed at a fixed location, such as roadside. The control circuitry **214** may be configured to activate the sensing function **218** to sense a surrounding area of the repeater device **302**. In an implementation, the control circuitry **214** may be configured to activate the sensing function **218** automatically when a moving object, such as the UE **304**, is detected within a communication range. In another implementation, the control circuitry **214** may be configured to activate the sensing function **218** based on an activation signal received from the UE **304**. Due to inadequate data throughput and signal strength of a cellular connectivity directly from a base station, such as the first base station **108A**, the activation signal may be communicated by the UE **304** to the repeater device **302**, for example, via an out-of-band channel, such as a Wi-Fi or other personal area network communication channel. The UE **304** may have a preinstalled application. Based on a defined setting, the preinstalled application may cause the UE **304** to send the activation signal automatically to the repeater device **302** when the UE **304** moves within a communication range of the repeater device **302**. Alternatively, based on user input to an application interface of the preinstalled application, the UE **304** may send the activation signal to the repeater device **302**. In yet another implementation, the central cloud server **102** may be communicatively coupled to the preinstalled application of the UE **304** and may detect the position of the UE **304**, and thereafter may direct the application to cause to the UE **304** to communicate the activation signal to the repeater device **302**.

[0064] The repeater device **302** may be a dual-function edge device in which a first portion **306A** of the first antenna array **306** is used for the sensing. The control circuitry **214** may be further configured to recognize the UE **304** in motion to be a valid device to receive one or more services from the repeater device **302** by use of the first portion **306A** of the first antenna array **306**. The control circuitry **214** may be further configured to track a position of the UE **304** with a centimeter-level accuracy (i.e., less than 5 or 10 cm of positioning error) in motion from the repeater device **302** by use of the first portion **306A** of the first antenna array **306**. For example, a RF wave **310** may be communicated to sense and track the position of the UE **304**. The control circuitry **214** may be further configured to execute beamforming to direct a first beam of RF signal **312** having a signal strength greater than a threshold to the UE **304** (i.e., stronger signal is concentrated on the UE **304**) in motion that is recognized as the valid device, based on the track of the position of the UE **304**. The first beam of RF signal **312** may be communicated in a radiation pattern (i.e., a defined shape, such as a narrow or pencil beam) from the same antenna array, i.e., one or more second portions of the first antenna array **306**, while the sensing is executed concomitantly by the

first portion **306A** of the first antenna array **306**. Thus, the UE **304** may be able to execute uplink and downlink communication via the repeater device **302** with a comparatively higher data throughput rate based on the concentrated signal continuously received from the repeater device **302** even though the UE **304** rapidly changes its position and orientation while in motion along a travel path. Furthermore, the repeater device **302** that acts as an edge device is independent of the plurality of different WCNs **110** and thus can provide services to the UE **304** either in the first WCN **110A** or the second WCN **110B** as per choice or a current subscription of the UE **304** to a particular service provider to ensure seamless connectivity and increase QoE.

[0065] In accordance with another exemplary aspect, the repeater device **302** may not be deployed at a fixed location, and may be instead installed at a vehicle, and act as an edge device. The UE **304** may also be present within the vehicle. Thus, in this exemplary aspect, the repeater device **302** and the UE **304** may be co-located at the vehicle; however, the repeater device **302** may be wirelessly connected to the central cloud server **102** over an LTE control channel independent of the UE **304**. Alternatively stated, the repeater device **302** is communicatively coupled to the central cloud server **102** irrespective of the connectivity of the UE **304**. Further, in this exemplary aspect, the central cloud server **102** may be configured to predict a travel path of the repeater device **302** in motion (that moves along the vehicle in this case). In other words, the central cloud server **102** predicts travel path that the UE **304** is likely to take based on sensing information received from the repeater device **302**. The repeater device **302** may further include a sensor, for example, a geospatial position sensor (such as GPS sensor), which may capture velocity of its movement. This captured sensing information, for example, the velocity (e.g., indicating speed and moving direction) is used by the central cloud server **102** to determine one or more alternative wireless connectivity options that may be made available to the UE **304** in motion. In other words, the central cloud server **102** may be guided by the velocity information, which in turn may trigger the central cloud server **102** to elastically alter how many directives (or instructions for alternative wireless connectivity options) it queues to the repeater device **302**. The central cloud server **102** may be configured to pre-load impending choices in terms of one or more alternative wireless connectivity options to minimize signaling latency between the repeater device **302** and a base station and/or the central cloud server **102** (In this case the UE **304** may be in a vehicle where the repeater device **302** is co-located as an edge device within a vehicle). In other words, the central cloud server **102** may be configured to communicate one or more alternative wireless connectivity options to the repeater device **302** (i.e., an edge device), where the one or more alternative wireless connectivity options are used by the UE **304** (which may be communicatively coupled to the repeater device **302**) as fallback options to maintain consistent wireless connectivity to a base station or the repeater device **302** (i.e., the edge device). As control channel at times can be lost briefly, such pre-loaded one or more alternative wireless connectivity options may be used by the UE **304** as guidance when cellular connectivity (e.g., 5G wireless connection) is lost for several seconds, such as when the UE **304** and the repeater device **302** provided in the vehicle moves in a tunnel or in some remote areas where cellular coverage is sparse. In an implementation, the prediction of the travel path of the repeater device **302** (and the UE **304** that is co-located) in motion may be executed based on the machine learning model of the central cloud server **102**.

[0066] In an implementation, the determined one or more alternative wireless connectivity options made available to the repeater device **302** as well as the UE **304** in motion comprises a plurality of different specific initial access information, where each of the plurality of different specific initial access information is capable to assist the repeater device **302** (e.g., an edge device) to bypass an initial access-search on the repeater device **302**. Each of the plurality of different specific initial access information are alternative options for wireless cellular connectivity communicated to the repeater device **302** based on current and upcoming position of the repeater device **302** along the predicted travel path. In an example, the central cloud server **102** may be configured to communicate two or more choices, say, four alternative wireless connectivity options, for example,

a first, second, third, and fourth choice for wireless connectivity so that the repeater device **302** can continue with mmWave connection with the least amount of service disruption. Alternatively stated, when the primary choice fails (i.e., the first communicated wireless connectivity enhanced information that includes a first specific initial access information is not usable for some unforeseen reasons, like loss of signal in a tunnel), other alternative wireless connectivity options can be selected to maintain continuous 5G connectivity for enhanced QoE by the repeater device **302** as well as the UE **304** (which may be connected for uplink and downlink communication via the repeater device **302**). Thus, having more alternative wireless connectivity options act as a powerful technique to maintain consistent 5G connectivity irrespective of an internal beam acquisition process of the repeater device **302**. Moreover, as such alternative wireless connectivity options comprises specific initial access information, the standard beam acquisition process is shortened, i.e., the time to scan and acquire new initial access information is shortened, and consequently, failure detection and recovering from it using the provided multiple alternative wireless connectivity options comprises plays a prominent role in increasing the QoE. Furthermore, as the repeater device **302** also communicates the sensing information including the moving direction of the UE **304**, a time-of-day, local traffic information, local road information, local construction information, local traffic light information, the prediction of the travel path can be made more accurately with more accurate failure detection and recovering options.

[0067] FIGS. **4A**, **4B**, and **4C**, collectively, is a flowchart that illustrates an exemplary method for accelerating user equipment (UE) specific beamforming for high performance and reliable communication, in accordance with an embodiment of the disclosure. FIGS. **4A**, **4B**, and **4C**, are explained in conjunction with elements from FIGS. **1**, **2A**, **2B**, **2C**, and **3**. With reference to FIGS. **4A**, **4B**, and **4C**, there is shown a flowchart **400** comprising exemplary operations **402** through **434**. The operations of the method depicted in the flowchart **400** may be implemented in the edge device **104A** (FIG. **2A**).

[0068] At **402**, a sensing function **218** may be activated to sense a surrounding area of the edge device **104A**. The control circuitry **214** may be configured to activate a sensing function **218** to sense the surrounding area of the edge device **104A**.

[0069] At **404**, a three-dimensional (3D) environment representation of the surrounding area of the edge device **104A** may be generated. The control circuitry **214** may be further configured to generate the 3D environment representation of the surrounding area of the edge device **104A**.

[0070] At **406**, the first UE **106A** in motion may be recognized to be a valid device to receive one or more services from the edge device **104A**. The control circuitry **214** may be further configured to recognize the first UE **106A** in motion to be a valid device.

[0071] At **408**, a position of the first UE **106A** in motion may be tracked from the edge device **104A**. The control circuitry **214** may be further configured to track a position of the first UE **106A** in motion from the edge device **104A**.

[0072] At **410**, beamforming may be executed to direct a first beam of RF signal having a signal strength greater than a threshold to the first UE **106A** in motion that is recognized as the valid device, based on the track of the position of the first UE **106A** and the generated 3D environment representation. The control circuitry **214** may be further configured to execute the beamforming function **220** to perform beamforming from the first antenna array **210A**, where the sensing may be executed in a first frequency that is different from a second frequency used to communicate the first beam of RF signal in a defined radiation pattern. In one implementation, a portion of the first antenna array **210A** is used for the sensing while one or more second portions of the first antenna array are used for the beamforming to direct the first beam of RF signal in a defined radiation pattern. Alternatively, in another implementation, the sensing radar **222** may be provided to perform the sensing in order to sense the surrounding area of the edge device **104A**.

[0073] At **412**, location coordinates of a plurality of reflective objects in the surrounding area of the edge device **104A** may be determined. The control circuitry **214** may be further configured to

determine the location coordinates of the plurality of reflective objects in the surrounding area of the edge device **104A**.

[0074] At **414**, the determined location coordinates of the plurality of reflective objects may be utilized to correlate a radiation pattern of the first antenna array **210A** to the plurality of reflective objects for improved directivity of communicated beams of RF signals. The control circuitry **214** may be further configured to utilize the determined location coordinates of the plurality of reflective objects to correlate the radiation pattern of the first antenna array **210A** to the plurality of reflective objects.

[0075] At **416**, a distance and an angle of the edge device **104A** from each of a plurality of mobile and stationary objects surrounding the edge device **104A** may be determined. The control circuitry **214** may be further configured to determine the distance and the angle of the edge device **104A** from each of the plurality of mobile and stationary objects surrounding the edge device **104A**.

[0076] At **418**, local traffic information may be determined in real-time or near real-time based on the sensed surrounding area of the edge device **104A** and the generated 3D environment representation of the surrounding area of the edge device **104A**. The control circuitry **214** may be further configured to determine the local traffic information in real-time or near real-time.

[0077] At **420**, a start time and an end time of a signal blockage may be predicted for the first UE **106A** in motion being serviced by the edge device **104A** from a second moving object based on a track of the second moving object in the surrounding area of the edge device **104A**. The control circuitry **214** predicts the start time and the end time of the signal blockage for the first UE **106A** in motion being serviced by the edge device **104A** from the second moving object. The control moves to **422A** or **422B** based on a defined setting at the edge device **104A**.

[0078] At **422A**, an alert of the predicted signal blockage for the first UE **106A** may be communicated to the central cloud server **102** along with the predicted start time and the end time of the signal blockage to cause the central cloud server **102** to instruct another edge device of the plurality of edge devices **104** with a specific initial access information to continue servicing the first UE **106A**. The control circuitry **214** may be further configured to communicate the alert to the central cloud server **102**.

[0079] At **422B**, an alert of the predicted signal blockage for the first UE **106A** may be communicated to another edge device of the plurality of edge devices **104** along with at least the predicted start time of the signal blockage and a specific initial access information to cause the other edge device to continue servicing the first UE **106A**. The control circuitry **214** may be further configured to communicate the alert of the predicted signal blockage to another edge device of the plurality of edge devices **104**.

[0080] At **424**, a second UE **106B** in motion may be recognized to be a valid device to receive one or more services from the edge device **104A** concomitantly with the first UE **106A**. The control circuitry **214** may be further configured to recognize the second UE **106B** in motion to be the valid device to receive one or more services from the edge device **104A** concomitantly with the first UE **106A**.

[0081] At **426**, a position of the second UE **106B** in motion may be tracked concurrently with the first UE **106A** from the edge device **104A**. The control circuitry **214** may be further configured to track the position of the second UE **106B** in motion concurrently with the first UE **106A** from the edge device **104A**.

[0082] At **428**, the radiation pattern of the first antenna array **210A** may be updated such that a second beam of RF signal having the signal strength greater than the threshold is concentrated on the second UE **106B** in motion concomitant to the first beam of RF signal that is concentrated on the first UE **106A** in motion. The control circuitry **214** may be further configured to update the radiation pattern.

[0083] At **430**, sensing information may be periodically communicated to the central cloud server **102** based on the sensed surrounding area of the edge device **104A**. The control circuitry **214** may

be further configured to set an offline mode or a connected mode at the edge device **104A**, wherein in the connected mode, the control circuitry **214** may be further configured to periodically communicate sensing information to the central cloud server **102** based on the sensed surrounding area of the edge device **104A**. The sensing information may comprise a distance and an angle of the edge device **104A** from each of a plurality of objects surrounding the edge device **104A**, a position of the edge device **104A**, a location and a moving direction of a plurality of UEs including the first UE **106A**, a time-of-day, local traffic information, local road information, local construction information, a local traffic light information, and local weather information. The central cloud server **102** may be configured to predict a travel path of the edge device **104A** (e.g., the repeater device **302**) and the first UE **106A** in motion when both the edge device **104A** and the first UE **106A** are co-located in a vehicle. The central cloud server **102** may be further configured to communicate one or more alternative wireless connectivity options to the edge device **104A** (e.g., the repeater device **302**) where the one or more alternative wireless connectivity options are used by the edge device **104A** as well as the first UE **106A** as fallback options to maintain consistent wireless connectivity to a base station in a case where the primary connection is lost. An example of the exemplary aspect related to the alternative wireless connectivity options, has been described in FIG. 3

[0084] At **432**, wireless connectivity enhanced information may be obtained from the central cloud server **102** based on the position of the edge device **104A** in the connected mode. The control circuitry **214** may be further configured to obtain the wireless connectivity enhanced information via the one or more donor antenna arrays **206** from the central cloud server **102** based on a position of the edge device **104A** in the connected mode. The wireless connectivity enhanced information may include the specific initial access information to bypass an initial access-search on the edge device **104A**.

[0085] At **434**, a response may be received from the central cloud server **102** that no handover is required for the edge device **104A** for one or more of the plurality of UEs based on the communicated sensing information. The control circuitry **214** may be further configured to receive the response from the central cloud server **102** that no handover is required for the edge device **104A** for one or more of the plurality of UEs based on the communicated sensing information.

[0086] FIG. 5 is a flowchart that illustrates an exemplary method for accelerating user equipment (UE) specific beamforming for high performance and reliable communication, in accordance with another embodiment of the disclosure. FIG. 5 is explained in conjunction with elements from FIGS. 1, 2A, 2B, 2C, and 3. With reference to FIG. 5, there is shown a flowchart **500** comprising exemplary operations **502** through **508**. The operations of the method depicted in the flowchart **500** may be implemented in the edge device **104A** (FIG. 2A).

[0087] At **502**, a sensing function **218** may be activated to sense a surrounding area of the edge device **104A**. The control circuitry **214** may be configured to activate a sensing function **218** to sense the surrounding area of the edge device **104A**.

[0088] At **504**, the first UE **106A** in motion may be recognized to be a valid device to receive one or more services from the edge device **104A**. The control circuitry **214** may be further configured to recognize the first UE **106A** in motion to be a valid device.

[0089] At **506**, a position of the first UE **106A** in motion may be tracked from the edge device **104A**. The control circuitry **214** may be further configured to track a position of the first UE **106A** in motion from the edge device **104A**.

[0090] At **508**, beamforming may be executed to direct a first beam of RF signal having a signal strength greater than a threshold to the first UE **106A** in motion that is recognized as the valid device, based on the track of the position of the first UE **106A**. The edge device **104A** may be a dual-function edge device in which a first portion of the first antenna array **210A** is used for the sensing and one or more second portions of the first antenna array **210A** is used for the beamforming for forming the first beam of RF signal in a defined radiation pattern.

[0091] Various embodiments of the disclosure may provide a non-transitory computer-readable medium having stored thereon computer-implemented instructions that, when executed by a computer, causes a communication apparatus to execute operations that include activating a sensing function to sense a surrounding area of the edge device **104A**. The operations further include generating a three-dimensional (3D) environment representation of the surrounding area of the edge device **104A**. The operations further include recognizing a first UE **106A** in motion to be a valid device to receive one or more services from the edge device **104A**. The operations further tracking a position of the first UE **106A** in motion from the edge device **106A**. The operations further include executing beamforming to direct a first beam of radio frequency (RF) signal having a signal strength greater than a threshold to the first UE **106A** in motion that is recognized as the valid device, based on the track of the position of the first UE **106A** and the generated 3D environment representation.

[0092] Various embodiments of the disclosure may provide a non-transitory computer-readable medium having stored thereon, computer-implemented instructions that, when executed by a computer, causes the computer to execute operations that include activating the sensing function **218** to sense the surrounding area of the edge device **104A** and recognizing the first UE **106A** in motion to be a valid device to receive one or more services from the edge device **104A**. The operations further include tracking a position of the first UE **106A** in motion from the edge device **104A**. The operations further include executing beamforming to direct a first beam of RF signal having a signal strength greater than a threshold to the first UE **106A** in motion that is recognized as the valid device, based on the track of the position of the first UE **106A**.

[0093] While various embodiments described in the present disclosure have been described above, it should be understood that such embodiments have been presented by way of example and not limitation. It is to be understood that various changes in form and detail can be made therein without departing from the scope of the present disclosure. In addition to using hardware (e.g., within or coupled to a central processing unit (“CPU”), microprocessor, microcontroller, digital signal processor, processor core, system on chip (“SOC”) or any other device), implementations may also be embodied in software (e.g., computer-readable code, program code, and/or instructions disposed of in any form, such as source, object or machine language) disposed for example in a non-transitory computer-readable medium configured to store the software. Such software can enable, for example, the function, fabrication, modeling, simulation, description, and/or testing of the apparatus and methods described herein. For example, this can be accomplished using general program languages (e.g., C, C++), hardware description languages (HDL) including Verilog HDL, VHDL, and so on, or other available programs. Such software can be disposed of in any known non-transitory computer-readable medium, such as semiconductor, magnetic disc, or optical disc (e.g., CD-ROM, DVD-ROM, etc.). The software can also be disposed of as computer data embodied in a non-transitory computer-readable transmission medium (e.g., solid-state memory any other non-transitory medium including digital, optical, analog-based medium, such as removable storage media). Embodiments of the present disclosure may include methods of providing the apparatus described herein by providing software describing the apparatus and subsequently transmitting the software as a computer data signal over a communication network including the internet and intranets.

[0094] It is to be further understood that the system described herein may be included in a semiconductor intellectual property core, such as a microcontroller (e.g., embodied in HDL) and transformed to hardware in the production of integrated circuits. Additionally, the system described herein may be embodied as a combination of hardware and software. Thus, the present disclosure should not be limited by any of the above-described exemplary embodiments but should be defined only in accordance with the following claims and their equivalents.

Claims

1. An edge device, comprising: an antenna array; and control circuitry configured to: track a position of user equipment (UE) in motion in terms of a current distance and a current angle from the edge device; execute beamforming to communicate a beam of radio frequency (RF) signal to the UE, wherein the beam of RF signal is communicated in a specific radiation pattern based on the track of the position of the UE, and the beam of RF signal has a signal strength greater than a threshold; and calibrate the specific radiation pattern of the antenna array to further communicate the beam of RF signal in a specific beam shape that is suitable for a defined period.
2. The edge device according to claim 1, wherein the edge device and the UE are co-located in a vehicle that is in motion.
3. The edge device according to claim 1, wherein the edge device corresponds to a repeater device at roadside and the UE is a vehicle in motion.
4. The edge device according to claim 1, wherein the control circuitry is further configured to: receive a connection request from the UE associated with a Wireless Carrier Network (WCN) of a service provider, wherein the connection request is received via an out-of-band communication; and recognize the UE as a valid device based on the connection request, wherein the beam of RF signal is communicated to the UE that is recognized as the valid device.
5. The edge device according to claim 1, wherein the control circuitry is further configured to: receive an authentication key from the UE based on the UE that is in a communication range of the edge device; and validate the authentication key to recognize the UE as a valid device, wherein the beam of RF signal is communicated to the UE that is recognized as the valid device.
6. The edge device according to claim 1, wherein the antenna array is a common point of reference to track the position of the UE as well as to communicate the beam of RF signal to the UE.
7. The edge device according to claim 1, wherein the control circuitry is further configured to activate a sensing function to sense a surrounding area of the edge device based on the UE that is in motion within a communication range of the edge device, and wherein the position is tracked based on the sense of the surrounding area.
8. The edge device according to claim 7, wherein the sense of the surrounding area is performed in a first frequency that is different from a second frequency used to communicate the beam of RF signal in the specific radiation pattern.
9. The edge device according to claim 7, wherein the edge device is a dual-function edge device, and a first portion of the antenna array is used for the sense and one or more second portions of the antenna array are used for the beamforming to communicate the beam of RF signal in the specific radiation pattern.
10. The edge device according to claim 1, wherein the control circuitry is further configured to: generate a three-dimensional (3D) environment representation of a surrounding area of the edge device, wherein the 3D environment representation is in a form of a point cloud, the 3D environment representation provides a reference to correlate between the tracked position of the UE and the point cloud that belongs to the UE, and the beam of RF signal is communicated further based on the generated 3D environment representation.
11. The edge device according to claim 1, wherein the calibration is based on changes in the tracked position of the UE, and the calibration dynamically alters a direction of the beam of RF signal, an angle of the beam of RF signal, and the specific beam shape of the beam of RF signal.
12. The edge device according to claim 1, wherein the beam of RF signal is a pencil beam.
13. The edge device according to claim 1, wherein the beam of RF signal is a narrow beam with a beam angle greater than that of a pencil beam.
14. A method for User Equipment (UE)-specific beamforming, the method comprising: tracking, by an edge device, a position of a UE in motion in terms of a current distance and a current angle from

the edge device; executing, by the edge device, beamforming to communicate a beam of radio frequency (RF) signal to the UE, wherein the beam of RF signal is communicated in a specific radiation pattern based on the track of the position of the UE, and the beam of RF signal has a signal strength greater than a threshold; and calibrating, by the edge device, the specific radiation pattern of an antenna array of the edge device to further communicate the beam of RF signal in a specific beam shape that is suitable for a defined period.

15. The method according to claim 14, wherein the edge device and the UE are co-located in a vehicle that is in motion.

16. The method according to claim 14, wherein the edge device corresponds to a repeater device at roadside and the UE is a vehicle in motion.

17. The method according to claim 14, further comprising sensing, by the edge device, a surrounding area of the edge device based on the UE that is in motion and within a communication range of the edge device, wherein the position is tracked based on the sensing of the surrounding area, and the sensing of the surrounding area is performed in a first frequency that is different from a second frequency used to communicate the beam of RF signal in the specific radiation pattern.

18. The method according to claim 14, further comprising: generating, by the edge device, a three-dimensional (3D) environment representation of a surrounding area of the edge device, wherein the 3D environment representation is in a form of a point cloud, the 3D environment representation provides a reference to correlate between the tracked position of the UE and the point cloud that belongs to the UE, and the beam of RF signal is communicated further based on the generated 3D environment representation.

19. The method according to claim 14, wherein the calibration is based on changes in the tracked position of the UE, and the calibration dynamically alters a direction of the beam of RF signal, an angle of the beam of RF signal, and the specific beam shape of the beam of RF signal.

20. The edge device according to claim 1, wherein the beam of RF signal is a pencil beam or a narrow beam with a beam angle greater than that of a pencil beam.
